

flow in patchy habitats, providing a framework for the conservation of species living in a network of habitat fragments and reserves.

CONCEPT CHECK 53.5

1. Describe three attributes of habitat patches that could affect population density and rates of immigration and emigration.
2. **DRAW IT** Circle the portion of the red curve in Figure 53.9 in which density-dependent population regulation would likely have the most pronounced effects.
3. **WHAT IF?** Suppose you were studying a species that has a population cycle of about ten years. How long would you need to study the species to determine if its population size were declining? Explain.
4. **MAKE CONNECTIONS** Negative feedback is a process that regulates biological systems (see Concept 40.2). Explain how the density-dependent birth rate of dune fescue grass exemplifies negative feedback.

For suggested answers, see Appendix A.

CONCEPT 53.6

The human population is no longer growing exponentially but is still increasing extremely rapidly

In the last few centuries, the human population has grown at an unprecedented rate, more like the elephant population in Kruger National Park (see Figure 53.8) than the fluctuating populations we considered in Concept 53.5. No population can grow indefinitely, however. In this section of the chapter, we'll apply the concepts of population dynamics to the specific case of the human population.

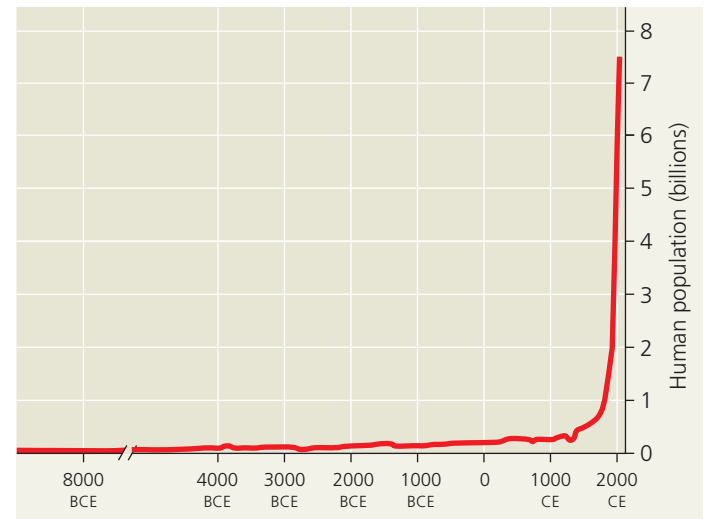
The Global Human Population

The human population has grown explosively over the last four centuries (**Figure 53.21**). In 1650, about 500 million people inhabited Earth. Our population doubled to 1 billion within the next two centuries, doubled again to 2 billion by 1930, and doubled still again to 4 billion by 1975. Notice that the time it took our population to double in size decreased from 200 years in 1650 to just 45 years in 1930. Thus, historically our population has grown even *faster* than exponential growth, which has a constant rate of increase and hence a constant doubling time.

The global population is now more than 7.6 billion people and is increasing by about 80 million each year. This translates into more than 200,000 people each day, the equivalent of adding a city the size of Amarillo, Texas. At this rate, it takes only about four years to add the equivalent of another United States to the world population. Ecologists

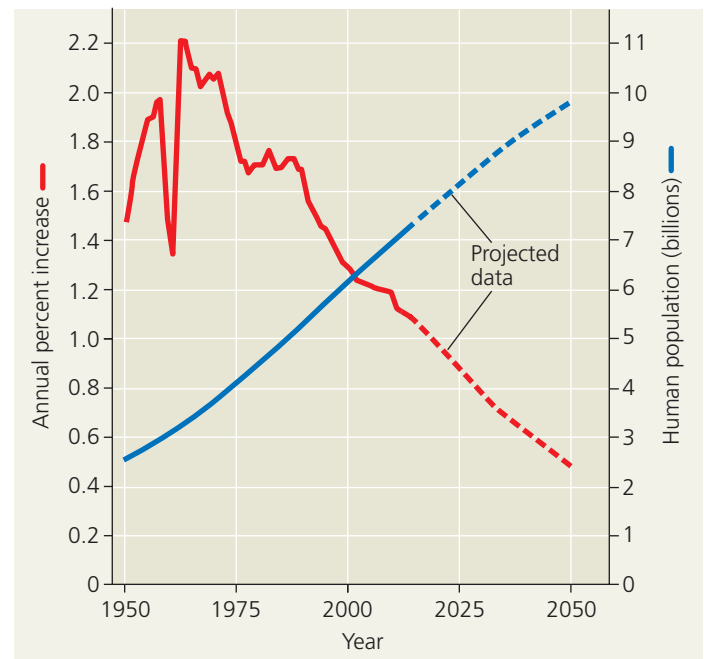
▼ Figure 53.21 Human population growth (data as of 2018).

The global human population has grown almost continuously throughout history, but it skyrocketed after the Industrial Revolution.



▼ Figure 53.22 Annual percent increase and recent human population growth.

The sharp dip in the rate of increase around 1960 is due mainly to a famine in China in which about 60 million people died. (Data as of 2018.)



VISUAL SKILLS The blue curve in this figure appears to tell a different story about human population growth than does the curve in Figure 53.21, yet both curves are accurate. Summarize what each curve conveys about recent human population growth and explain how both can be accurate.

predict there will be about 9.8 billion people on Earth by the year 2050.

Though the global population is still growing, the *rate* of growth began to slow during the 1960s (**Figure 53.22**). The annual rate of increase in the global population peaked at 2.2% in 1962 but was only 1.1% in 2018. Current models